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MILITARY SPECIFICATION

FIXING BATH, CONVENTIONAL PHOTOGRAPHIC, ACID HARDENING

This specification is approved for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 This specification covers the performance requirements of general purpose acid hardening baths for fixing and hardening of photographic emulsions. The sodium thiosulfate fixer will be provided as a single homogeneous powder mix. The ammonium thiosulfate fixer will be provided as a liquid concentrate which may be in one or two parts.

2. APPLICABLE DOCUMENTS

*2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

O-C-275	Chemicals, Photographic "Single Substance," General Specification For
MMM-A-178	Adhesive, Paper Label, Water-Resistant
PPP-B-636	Box, Shipping, Fiberboard
PPP-B-640	Boxes, Fiberboard, Corrugated, Triple-Wall
PPP-C-96	Cans, Metal, 28 Gage and Lighter

Military

MIL-B-26701	Bottles, Screw Cap and Carboys; Polyethylene Plastic

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 6750

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*2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American National Standards Institute

ANSI PH 4.11 Determing the Melting Point of a Nonsupport Layer of Films, Plates, and Papers, Method for
(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018).

3. REQUIREMENTS

3.1 Chemicals. The chemicals used shall be photographic grade conforming to O-C-275.

3.2 Prepared mixes

3.2.1 Dry powder mixes. The dry powders shall be homogeneously blended, free from lumps, free-flowing, readily soluble, and free from any foreign matter when dispensed into the unit containers.

3.2.2 Liquid concentrate mixes. The chemicals in the liquid concentrate shall be in solution. The liquid shall be free from any contaminating foreign matter when dispensed into the unit container or containers. When the concentrate is supplied as a two-part mix, the two parts shall be packed for shipment in a manner that will insure simultaneous delivery of both parts.

3.3 Storage

*3.3.1 Powder. The dry mixed powder, in the unit container, shall withstand storage as specified in 4.6.1.1 with no evidence of sulfurization or discoloration. The powder shall remain free-flowing, except for possible soft lumps or soft caking, and shall be readily soluble. No rust shall be observed in the interior of the container. The contents of the containers shall retain performance requirements capability as specified in Section 3.

*3.3.2 Liquid concentrate. The liquid concentrate, in unit container or containers, shall withstand storage as specified in 4.6.1.2 without physical evidence of sulfurization, precipitation, or discoloration. The contents of the containers shall retain performance requirements capability as specified in Section 3.

*3.3.3 Liquid concentrate cold storage. The liquid concentrate in one or two-part containers shall be stored as specified in 4.6.1.3. Any evidence of crystallization at the end of the 24 hour time period will be cause for rejection.

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*3.4 Dissolving and appearance of solution. The freshly mixed fixing bath, when tested in accordance with 4.6.2 shall be clear and colorless and have no more than a slight flocculence. The mixing process shall not exceed 10 minutes.

*3.5 Acidity. The hydrogen ion concentration (pH) of freshly mixed fixing bath, when tested in accordance with 4.6.2 shall be within 0.4 pH units of the pH of Standard Solution I or II, as appropriate (see 4.5).

*3.6 Sludging. When tested as specified in 4.6.4, the sludging characteristics of unused, working fixer solutions shall be as follows:

a. Dry powder mix: A working solution prepared from the dry powder mix shall develop no sludge when tested as specified in 4.6.4.a.

b. Liquid concentrate: A working solution prepared from the liquid concentrate shall develop no sludge when tested as specified in 4.6.4.b.

*3.7 Buffering capacity. When tested as specified in 4.6.5, the amount of potassium hydroxide required for the test solution will not be less than that required for the appropriate standard solution (see 4.5).

*3.8 Clearing time. When tested as specified in 4.6.6, the time required to clear film processed in the fixing bath shall not be greater than 20 percent longer than the time required to clear the same film processed in standard solution No. I or No. II, whichever is applicable (see 4.5).

*3.9 Clearing capacity. When tested as specified in 4.6.7, the clearing capacity of the fixing bath shall not be less than 80 percent of the total square inch clearing capacity of the standard solution No. I or No. II, whichever is applicable (see 4.5).

*3.10 Thiosulfate content. When tested as specified in 4.6.8, the quantity of thiosulfate ion in the fixing bath shall not be less than 90 percent of the thiosulfate ion content of Standard Solution No. I or No. II, whichever is applicable (see 4.5).

*3.11 Hardening action. When tested as specified in 4.6.9, the film emulsion fixed in the bath shall not melt, reticulate, or show any evidence of degradation at any temperature below the melting point of the strip fixed in the Standard Solution No. I or II, whichever is applicable (see 4.5).

*3.12 Sulfurization. When tested as specified in 4.6.10, the fixing bath shall show no precipitation other than a slight flocculence.

3.13 Unit size. The chemicals shall be packaged in unit sizes sufficient to prepare working fixing baths as specified by the procuring activity (see 6.2).

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3.14 Net content.

*3.14.1 Net weight of dry mixed powder. The net weight of the powder in each unit container shall be within ±3 percent of the marked weight.

*3.14.2 Net volume of liquid concentrate. The net volume of the liquid in each unit container shall be within ±3 percent of the marked volume.

*3.15 Leakage. The product shall be packaged in containers sufficient to prevent leakage.

*3.15.1 Dry powder mix. When tested as specified in 4.6.11.1, there shall be no rusting, lumping, moisture, or other evidence of container leakage.

*3.15.2 Liquid concentrate. When tested as specified in 4.6.11.2, there shall be no crystal formation around the cap or other indication of container leakage.

*3.16 Workmanship. Workmanship shall be in accordance with high grade commercial practice covering the manufacture and packaging of photographic materials. The manufacturer's commercial warranty will apply, and will include, as a minimum, replacement of products not suitable for the intended purpose.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection. The examinations and tests comprising quality conformance inspection are classified as specified in Table II.

*4.2.1 Sampling for inspection and acceptance. Sampling for inspection and acceptance shall be in accordance with MIL-STD-105, except where otherwise indicated. For purposes of sampling, an inspection lot shall consist of all material of the same type and item number submitted for inspection and delivery at one time.

4.3 Examination of the end item. Examination of the end item shall be made in accordance with the classification of defects set forth below. The lot size, for purposes of determining sample

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size, shall be expressed in units of filled primary containers for examination in 4.3.1, 4.3.2 and 4.3.3 and in units of shipping containers for examination in 4.3.4. The inspection level and acceptable quality level (AQL), expressed in defects per 100 units, shall be as indicated in Table II.

4.3.1 Examination of the end item container for defects in closure, appearance, construction, workmanship and marking. The sample unit for this examination shall be one filled primary container. The end item container shall be examined for defects specified in Table I.

TABLE I

EXAMINE	DEFECT
Closure	Any leakage or seepage of contents
	Closure not as specified, not hermetically sealed
Appearance, construction and workmanship	Cracked, crushed, or any other defect affecting serviceability. Not in container specified. Not clean, any evidence of dirt, rust, dust or foreign matter.
Label data	Directions for mixing and directions for use omitted, illegible, incorrect, incomplete, or not in accordance with contract requirements.

4.3.2 Examination of the end item container for defects in appearance of contents and for space utilization. The sample unit for this examination shall be one filled primary container. The end item container shall be examined for the defects specified in Table III.

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TABLE II EXAMINATION AND TESTS

EXAMINATION/TEST	Spec Reference		Reqmts apply Indiv Unit	Determin per Smpl Unit	Rpt Results as		Insp Level	AQL
	Req Para	Insp Para			Pass/ Fail1/	Numerically to Nearest 2/		
End item container:								
Closure, appearance, etc		4.3.1	yes		X		I	2.5
Contents, space utilization		4.3.2	yes		X		S-3	1.5
Net contents		4.3.3	yes		X		S-2	2.5
Preparation for delivery		4.3.4	yes		X		S-2	4.0
Storage:	3.3	4.6.1						
Condition of fixer and presence of rust as applicable			yes	one	X		S-13/	2.5
Dissolving test	3.4	4.6.2	yes	one	X		S-13/	2.5
Acidity	3.5	4.6.3	yes	avg two		0.1 pH	S-13/	2.5
Sludging	3.6	4.6.4						
Sludge			yes	two	X		S-13/	2.5
Buffering capacity	3.7	4.6.5	yes	two	X		S-13/	2.5
Clearing time	3.8	4.6.6	yes	one		second	S-13/	2.5
Clearing capacity	3.9	4.6.7	yes	avg two		0.1 sq in	S-13/	2.5
Thiosulfate content	3.10	4.6.8	yes	avg two	X	0.1% iodine	S-13/	2.5
Hardening action	3.11	4.6.9						
Melting			yes	one		degree F	S-13/	2.5
Reticulation and degradation			yes	one	X		S-13/	2.5
Sulfurization	3.12	4.6.10	yes	one	X		S-13/	2.5
Leak Test	3.15	4.6.11	yes	one	X		S-14/	4.0

NOTES:

1/ If failure is indicated, report degree or numerical point of failure.
2/ Test reports also include all values on which results are based
3/ Not less than 7 sample units shall be tested
4/ Not less than 5 sample units shall be tested

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TABLE III

EXAMINE	DEFECTS
Appearance	Not homogeneous. Not in liquid form when required. Not in powder form when required. Any undissolved material in liquid concentrate. If powder form, contains lumps, is caked, or not free-flowing. Any discoloration; material contains dark specks or evidence of foreign matter.
Inside coating of powder container 1/	Missing, does not completely cover inner surface, flakes, chips, or not as specified. Rust on inside of container.
Space utilization of container	Less than 80 percent of internal space utilized; contents of container spills when opened in upright position.

1/ Contractor shall furnish a certificate of compliance that inner coating will not react physically or chemically with the contents of the container.

*4.3.3 Examination of end item container for defects in net contents. The sample unit for this examination shall be one filled primary container. When liquid concentrate is supplied as a two-part fixer, the sample unit shall consist of the two-part container. The net weight or volume of each sample unit shall not vary more than 3 percent from the specified quantity.

4.3.4 Examination of preparation for delivery. An examination shall be made to determine that the packaging, packing and markings comply with the requirements of Section 5 of this specification. The sample unit for this examination shall be one shipping container, fully packed, selected just prior to the closing operation. Shipping containers fully prepared for delivery shall be examined for closure defects. For dry mix powders, mixing and packaging shall be performed in a room with an absolute humidity of not more than 10 grams of water vapor per cubic meter of air. The defects for which the end item is examined are specified in Table IV.

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TABLE IV

EXAMINE	DEFECT
Packaging	Unit package not packaged as specified. Packaging material not as specified.
Packing	Not in accordance with contract requirements. Arrangement or number of unit containers or complete two-part containers per shipping container not in accordance with contract requirements. Container not as specified; closures not accomplished by specified or required method of materials. Any nonconforming component, component missing, damaged, or otherwise defective, affecting serviceability. Inadequate application of components, such as: Incomplete closure of case liners, container flaps, loose or inadequate strapping, bulged or distorted containers.
Count	Less than specified or indicated number of containers.
Weight	Gross weight exceeds specified requirements tolerance.
Marking	Interior or exterior markings (as applicable) omitted, illegible, incorrect, incomplete, or not in accordance with contract requirements.

4.4 Testing the end item. The end item shall be tested for the applicable characteristics as indicated in Table I for each lot presented for inspection for each designated fixer specified by the procuring activity. The sample unit for the leak test shall be one primary container of dry mix powder or one or two-part primary containers of liquid concentrate, as applicable. The sample unit for the remainder of the tests shall be one primary container of dry mix powder or one or two-part primary containers of liquid concentrate, as applicable. The inspection levels and AQLs, expressed in defects per 100 units, shall be as indicated in Table II.

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4.4.1 Test data. The contractor shall keep the following test data as specified herein, and a copy shall be made available to the Government inspector.

a. Test data shall be kept on each of the individual bulk chemicals used in the formula to insure that each chemical meets the chemical quality requirements. In lieu of these data, the contractor shall keep on file an average analysis for the manufacturer of each bulk chemical to prove that the chemicals supplied were in accordance with the specification.

b. Test data on each batch of given photographic processing formula to assure that it has been accurately compounded.

*4.5 Standard formulas. Table V lists the chemical makeup of three standard solutions which shall be used in determining the acceptability of a product in accordance with this specification. Standard Solution No. I shall be the standard of comparison for sodium thiosulfate formulations and Standard Solution No II shall be the standard of comparison for ammonium thiosulfate formulations. Standard Solution No III shall be used for the developer in those tests requiring a developer. These formulations are for standards only and shall not be packaged.

TABLE V

Dissolve all chemicals in the order given in distilled or deionized water.

Standard Solution No. I		
Sodium Thiosulfate Formula		
Water approx 51.6°c (125°F)	600.0	milliliters (ml)
Sodium thiosulfate crystals ($5H_20$)	240.0	grams (gm)
Sodium sulfite (anhydrous)	15.0	gm
Acetic acid (28 percent)	50.0	ml
Boric acid crystals	7.5	gm
Aluminum potassium sulfate	15.0	gm
Water, a quantity sufficient to make	1.0	liter
Standard Solution No. II		
Ammonium Thiosulfate Formula		
Water approx 32.2°c (90°F)	500.0	ml
Ammonium Thiosulfate (60 percent)	225.0	ml
Sodium Sulfite (anhydrous)	15.0	gm
Acetic acid (28 percent)	50.0	ml
Boric acid crystals	7.5	gm
Aluminum potassium sulfate	15.0	gm
Water, a quantity to make	1.0	liter

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TABLE V (cont)

Standard Solution No. III	
Water approx 32.2°c (90°F)	750.0 ml
p-Methylaminaphenol sulfate	2.5 gm
Hydroquinone	2.5 gm
Sodium sulfite (anhydrous)	30.0 gm
Sodium metaborate ($4H_2O$)	10.0 gm
Potassium bromide	0.5 gm
Water, a quantity to make	1.0 liter

4.6 Test methods

4.6.1 Storage test

*4.6.1.1 Dry mix powder. Sample units of dry-mix powder shall be stored in a constant temperature oven for 30 days at 37.8 ± 3° Celsius (C) (100° ± 5° Fahrenheit [F]). The container shall then be removed and allowed to cool to ambient room temperature for 2 hours. The container shall then be opened and the powder examined. Failure to meet the requirements specified in 3.3.1 shall be cause for rejection.

*4.6.1.2 Liquid concentrate. One or two part liquid concentrate containers shall be stored in a constant temperature oven for 30 days at 37.8 ± 3°c (100° ± 5°F). The containers shall then be removed and allowed to cool to ambient room temperature for 2 hours. The containers shall then be opened and the liquid examined. Failure to meet the requirements specified in 3.3.2 shall be cause for rejection.

*4.6.1.3 Cold storage for liquid concentrate. One or two part liquid concentrate selected from each lot shall be stored in a constant temperature environment of -17.8 ± 3°c (0° ± 5°F) for 24 hours. The containers shall then be removed and allowed to stand at room temperature not to exceed 23.9°c (75°F) for 24 hours. Failure to meet the requirements of 3.3.3 shall be cause for rejection.

*4.6.2 Dissolving and appearance. The chemicals obtained from the storage test, 4.6.1, shall be dissolved according to the manufacturer's recommended instructions to the volume specified in the contract for "quantity to make". The freshly mixed solution will be cooled to 21.1 ± 3°c (70° ± 5°F). Failure to meet the requirements specified in 3.4 shall be cause for rejection.

NOTE: The working fixer obtained from the dissolving test, if acceptable, shall be the working fixer used in conducting the subsequent tests specified herein.

4.6.3 Acidity. The acidity of the test solution and standard solution I or II, as appropriate, prepared in accordance with 4.5 and Table V, shall be determined with a properly calibrated pH

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meter. Any deviation greater than that allowed in 3.5 shall be cause for rejection.

*4.6.4 Sludging. Measure 100ml of working fixer solution, mixed as previously specified in 4.6.2, and pour into a 400ml beaker. The temperature shall be maintained at 20 ± 1°c (68° ± 2°F) for the powder formualtion and at 32.2 ± 1°C (90° ± 2°F) for the liquid concentrate formulation. While stirring rapidly by mechanical means, add 2 Normal (N) potassium hydroxide at the rate of 1.5ml per minute until the following conditions have been established:

a. The solution formulated from dry powder mix is adjusted to a pH of 6.0. The solution will be examined for evidence of sludging. Failure to meet the requirements of 3.6.a shall be cause for rejection.

b. The solution formulated from the liquid concentrate is adjusted to a pH of 7.0. The solution will be examined for evidence of sludging. Failure to meet the requirements of 3.6.b shall be cause for rejection.

*4.6.5 Buffering capacity. When tested in accordance with 4.6.4, the total amount of 2N potassium hydroxide added to either solution shall be evaluated against the amount of 2N potassium hydroxide required to obtain the same conditions with standard solution No. 1 or No. II, as applicable. Failure to meet the requirements specified in 3.7 shall be cause for rejection.

*4.6.6 Clearing time. Fill one 2.54 x 20.32 Centimeters (CM) (1x8 inch [in]) test tube with the freshly mixed bath solution prepared in the dissolving test. Fill another 2.54 x 20.32 (CM) (1x8in) test tube with standard solution No. I or No. II, whichever is applicable. Maintain standard solution No. I and its equivalent test solution at 20 ± 0.5°c (68° ± 1°F), or standard solution No. II and its equivalent test solution at 32.2 ± 0.5°c (90° ± 1°F). Fill another container with 2 liters (2000ml) of standard solution No. III. Cut 1.27 x 20.32cm (½in x 8in) film strips from a fresh piece of Eastman Kodak Plus -X pan sheet film, or equal. These unexposed strips shall be immersed in standards solution No. III for 3 minutes at 20°c (68°F), then rinsed for 60 seconds in running water, at ambient temperature, and drained for 5 seconds. Immerse a portion of these film strips in each of the fixing bath solutions, both test and standard, without agitation for 15 seconds. As the timing is started, immerse the remainder of the film strips and agitate continuously beneath the surface of each solution. Turn on room lights immediately after the strips are fully immersed in the solutions. The first immersed portion of the film aids in determining when strips are fully cleared. The disappearance of the line between the two sections is the clearing time. Failure to meet the requirements specified in 3.8 shall be cause for rejection.

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*4.6.7 Clearing capacity. Fill an 21.59 x 27.94cm (8½in x 11in) photographic film processing tray with 600ml of standard solution No. I or No. II, whichever is appropriate, at 20 ± 1°c (68° ± 2°F). Immerse and completely clear consecutive 20.32 x 25.40cm (8in x 10in) sheets of undeveloped Eastman Kodak Film -X pan sheet film, or equal. The last sheet that completely clears in 3 minutes shall consitute the clearing capacity. Calculate the total square inch clearing capacity. Fill another 21.59 x 27.94cm (8½in x 11in) photographic film processing tray with 600ml of the freshly mixed fixing bath solution, prepared in the dissolving test, at 20 ± 1°c (68° ± 2°F). Use the same procedure as with the standard solution and calculate the clearing capacity of the test solution. Failure to meet the requirements specified in 3.9 shall be cause for rejection.

*4.6.8 Thiosulfate content. Pour 100ml ice cold water into each of two 500ml Erlenmeyer flasks. Accurately pipet two (2) ml of working fixer into one and two (2) ml of standard solution No. I or No. II, whichever is applicable, into the other. Add two drops of bromothymol blue indicator to each and adjust to neutrality (green end point) with 1.0 N sodium hydroxide solution. Add 20ml of 37 percent formaldehyde to each flask and again adjust to neutrality by using 1.0 N hydrochloric acid, if color is blue, or 1.0 N sodium hydroxide, if color is yellow. Let stand 5 minutes and titrate with 1.0 iodine using a starch indicator. Record the amount of iodine used for each sample. The thiosulfate ion content is proportional to the amount of iodine required to effect a color change. Failure to meet the requirements of 3.10 shall be cause for rejection.

*4.6.9 Hardening properties. Place 1 liter (1000ml) of standard solution No. III in a container. Fill a test tube with 100ml of fresh fixing solution No. I or No. II, whichever is applicable. To this solution add 25ml of standard solution No. III. In total darkness, cut 1.27 x 20.32cm (½in x 8in) film strips from a fresh piece of Plus-X pan sheet film or equal. Using a high density straight edge, mask half of the test strip lengthwise. Expose to a light source to completely expose the unmasked half of the test strip lengthwise. A properly exposed strip will have a maximum density on one-half of the strip and no density, other than base density, on the other half after development and fixing process is completed. Place this film strip in standard solution No. III so that it is completely submerged. With occasional agitation, allow the strip to remain in the developer for 5 minutes at 20 ± 1°c (68° ± 2°F). Remove the strip from the developer and rinse under running water for 30 seconds. Immediately transfer the strip to the test tube containing the fixing bath solution. With occasional agitation, allow the film strip to remain in the fixing solution for a time equal to twice the normal clearing time as determined in 4.6.6. Transfer the film strip to a tray of running water and allow to wash for 10 minutes. The strip shall then be dried. Determine the melting point of the gelatin coating of the film strip in accordance with ANSI STD PH 4.11. Using 100ml of the freshly mixed fixing solution

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prepared in the dissolving test in place of the 100ml of standard solution No. I or No. II, repeat the above test under identical conditions. Test results shall be evaluated against the requirements. Failure to meet requirements specified in 3.11 shall be cause for rejection.

NOTE: To expedite testing, the above tests can be run concurrently provided the test film strips are adequately identified to the fixing solution used for each film strip.

*4.6.10 Sulfurization. Half-fill a 500ml glass stoppered bottle with fresh working solution of the test fixing bath. Place the closed bottle in an oven at a temperature of 43.3 ± 1.5°c (110° ± 3°F) for a period of 7 days. CAUTION: Do not insert the stopper tight enough to allow pressure to build-up inside the bottle. At the end of this period of time, examine the solution. Failure to meet the requirements specified in 3.12 shall be cause for rejection.

*4.6.11 Leak test

*4.6.11.1 Dry powder mix. The sealed container shall be immersed in water at a temperature of 13.3 to 29.4°c (56° to 85°F) for 24 hours with the uppermost part of the container at a depth of three feet. At the conclusion of the immersion period, the container shall be removed, dried, opened and the inside of the container and its contents examined for evidence of leakage. Failure of the container to meet the requirement specified in 3.15.1 shall be cause for rejection.

*4.6.11.2 Liquid concentrate. The sealed container shall be placed in a plastic bag and then put into storage at 60°c (140°F) for 24 hours. Failure of the container to meet the requirements specified in 3.15.2 shall be cause for rejection.

5. PACKAGING

5.1 Preservation and packaging. Preservation and packaging shall be Level A or C as specified (see 6.2).

5.1.1 Level A. Unless otherwise specified by the procuring activity, Level A preservation and packaging shall be as specified in 5.1.1.1 or 5.1.1.2.

5.1.1.1 Dry mixed powder. The dry mixed powder form of fixer shall be unit packaged, in "quantity to make" size as specified by the procuring activity, in containers conforming to PPP-C-96, Type I, Class I or Class 2 or Type V, Class 1, round, 0.113k (0.25 pounds) electrolytic tin plate. The exterior surfaces of the container shall be finished in accordance with Plan B.

NOTE: At the option of the procuring activity, the manufacturer's commercial container may be accepted, in lieu of containers finished in accordance with Plan B, provided this commercial container meets all other requirements specified, including the container marking requirements below.

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5.1.1.2 Liquid concentrate. The liquid concentrate form of fixer shall be packaged, in "quantity to make" size as specified by the procuring activity, in polyethylene bottles, of appropriate size, conforming to the quality specified in MIL-B-26701. Over package each unit container, or both containers if concentrate is supplied as a two-part mix, in a fiberboard box conforming to PPP-B-636, Style RSC, Type CF, Class weather resistant, grade V3c.

5.1.2 Level C. The fixer, either powder form or liquid concentrate form shall be preserved and packaged in accordance with manufacturer's commercial practice, provided the container meets quality requirements as specified herein.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. Fixer, preserved and packaged as specified in 5.1.1, shall be packed in fiberboard boxes conforming to PPP-B-636, style RSC, type CF, class weather resistant, grade V3c, or PPP-B-640, class 2, style E. Maximum weight of container and contents shall not exceed 113.4k (250 pounds).

5.2.2 Level B. Fixer, preserved and packaged as specified in 5.1.1, shall be packed in fiberboard boxes conforming to PPP-B-636, style RSC, type CF, class domestic variety SW, grade 275, or PPP-B-640, class 1, style E. Maximum weight of container and contents shall not exceed 113.4k (250 pounds).

5.2.3 Level C. Fixer, preserved and packaged as specified in 5.1.2, which requires overpacking for acceptance by the carrier shall be packed in exterior type shipping containers in such a manner that will insure safe transportation at the lowest rate to the point of delivery for immediate use. Containers shall meet Interstate Commerce Commission regulations of common carriers as applicable to the mode of transportation.

5.3 Container marking. The unit container shall be durably and legibly marked by a lithographic, silk screen, or similar process with a highly contrasting color or lettering directly on the body of the container provided the order is for at least 5,000 units. A paper label may be used for procurement of less than 5,000 units. Each label shall be coated on the back side with a waterproof adhesive. This waterproof adhesive shall be used to securely affix the paper labels to the container. The labels shall have waterproof black printing on a high grade white paper stock, and the adhesive for labels shall conform to MMM-A-178. When liquid concentrate is supplied as a two-part mix, each of the containers comprising the unit container shall be durably and legibly marked for identification.

5.3.1 Marking data. The container marking or printed label shall show the item identification, directions for mixing and using, manufacturer's name, date of manufacture, and the final volume, when mixed for use, in the following units:

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a. US gallons or fluid ounces
b. Imperial gallons or fluid ounces
c. Liters or milliliters

6. NOTES

6.1 Intended use. The fixers specified herein are intended for use in conventional tray, or tank or machine processing of photographic films and papers.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Quantity required.
c. Unit package size (expressed as "Quantity to make") (see 3.13).
d. Whether liquid or powder (see 1.1).
e. Selection of applicable levels of preservation and packaging, and packing.

6.3 International standardization. Certain provisions (5.3.1) of this specification are the subject of international standardization agreements (ASCC Air Std. 52/3). When amendment, revision, or cancellation of this specification is proposed which will effect or violate the international agreement concerned, the preparing activity will take appropriate reconcillation action through international standardization channels including departmental standardization offices, if required.

6.4 Marginal notations. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - EL
Navy - AS
Air Force - 99

Reviewers:
DGSC - GS

International Interest

Preparing Activity:
Air Force - 99
(Project No. 6750-0314)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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